SECTION A

MOTION AND FORCES

ITEM 1:

On one of the newly developed apartments, water supply systems need to be clearly fixed to ensure constant flow of clean water at right pressure and quantity. However, one of the clients on the second flow of the apartment complained about the flow pressure and volume flow rate of water entering the bathroom. They called a particular worker who claimed that he can fix any of the problems faced. He was told that water enters the apartment through a pipe with an inside diameter of 2.0 cm at absolute pressure of 4×10^5 Pa and speed of 2.0 ms^{-1}. The pipe leading to the second- floor bathroom 5.0 m above is 1.0 cm in diameter. The worker said that he can even use this information to establish the accuracy in the client's complaint but he failed and needed assistance.

The worker instead requested for a ladder to climb and establish if there are other water flow problems such as pipe problems. He was given a uniform ladder of weight 180N and length 4.0 m and he inclined it at 60^0 to the rough horizontal ground (coefficient of friction is 0.47) and the upper side leaned against a smooth vertical wall. When the worker of weight 800N had climbed some length on the way up the ladder, he feared that it would slip, he thus asked for help to establish how far he could climb just before the ladder could slip and how the coefficient of friction was determined.

As he was climbing, there was a Dstv satellite dish and the client requested him to adjust it and enable her to obtain clear signals of CNN - a USA Tv channel from communication satellites. The worker wondered how such satellites work to distribute signals globally.

The client then received a call from her professional plumber who said the he was rushing in his car to reach the apartment and fix any water challenges but he was worried that he may not reach in time since the road had many flat straight and curved sections.

Hint:

A good water supply system to the bathroom should have a pressure of at least 3.5×10^5 *Pa and a volume flow rate of at least 0.7 litres per second.*

Tasks: As a physics student,

a) Analyse the information to address the concerns of the hired worker.
b) Help the plumber understand road section designs he can develop to enable him travel at highest possible safe speeds.

ITEM 2:

While at the airport, a pilot conducts a final check on whether his aircraft with a total wing area of 40 m^2 can ably create sufficient lift to takeoff. Takeoff occurs when the lift exceed the total weight of the aircraft. Flight instruments indicate that air speed above the wings is 120 ms^{-1} while that below is 100ms^{-1}. He does not know how to interpret the measurements in relation to the shape of the aircraft wings to create a lift force. He also wants to know whether these speeds are sufficient to cause lift.

During the flight, the aircraft's communication and navigation system fails, and the pilot carries out an emergency evacuation of the passengers using a parachute. Just some few metres above the ground Musa's parachute developed a puncture and felt weightless as he descended. Upon reaching the ground, he flexed his knees while the other passengers reported of normal weight and did not flex their knees. Musa and those passengers narrated the situation to the rescue team who were waiting. The team did not understand this.

Dr. Sarah is tasked with deploying a 2000 kg emergency communication satellite to locate the wreckage of the aircraft and restore regional links. She aims for a stable orbit at a height of 800 km above the earth's surface. She has been advised that the mission will succeed only if the launch energy needed stays within their rocket's capacity of 6.5×10^{10}J. She needs evaluate the success of the mission, be guided on the launching procedures, understand what is likely to happen in case the propulsion system fails when it has just

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reached this altitude in order to be able to estimate when it likely to return to the earth's surface.

- Density of air $\rho_a = 1.29\ kgm^{-3}$
- Total mass of the aircraft = 1500kg
- Acceleration due to gravity at the earth's surface, $g = 9.81 ms^{-2}$.
- Radius of the earth, $R = 6.4 \times 10^6$ m

Tasks;

(a) Respond to the concerns of the pilot, the rescue team and establish whether the launch of new communication satellite will be successful.

(b) Guide Dr. Sarah to write launching procedures, circumstances under which launching may fail and what is likely to happen if it fails and the necessary estimates.

PARTICLES

ITEM 3:

An industrial manufacturing plant utilizes a modern diagnostic X-ray tube operating at 40 kV to maintain standard thickness gauge of aluminium sheets produced by a certain factory. The automated gauge uses a radioactive source that emits beta particles. The factory workers do not understand why beta emitting source is preferred to any other source. The existing hydroelectric power grid faces a daily deficit of 6.9×10^{13} J to run the factory. Therefore, a small-scale nuclear fission reactor has been set up to cater for the deficit. The workers in the factory do not know how nuclear fission releases energy, how such systems are regulated to maintain safety and why a neutron is used not any other particle such as alpha or a proton. In an effort to educate the workers, the engineer has provided a standard nuclear equation for the nuclear process:

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3^{1}_{0}n$$

Due to limited availability of uranium, the workers want to know whether 1 kg of pure uranium is enough to cater for the daily deficit energy.

While meeting a group of physics learners on study trip at the factory were notified that the quality of aluminium sheets relies on its interatomic spacing which is known to be 2.3×10^{-10} m. The aluminium sheet is of high quality if the diffracting angle for X-rays does not fall below 7° for the second order diffraction. The students are required to evaluate the quality of aluminium

sample sheets, provide detailed procedure of how the factory maintains uniform thickness of a given aluminium sheet.

- Mass of $^{235}_{92}U = 235.0439u$, Mass of $^{1}_{0}n = 1.0087u$,
- Mass of $^{141}_{56}Ba = 140.9144u$, Mass of $^{92}_{36}Kr = 91.9262u$, $1u = 931MeV$
- Speed of electromagnetic radiation in vacuum, $c = 3.0 \times 10^8 ms^{-1}$
- Plunk constant, $h = 6.6 \times 10^{-34} Js$.
- Electronic charge, $e = 1.6 \times 10^{-19} C$.
- Avogadro constant, $N_A = 6.02 \times 10^{23} mol^{-1}$

Tasks:

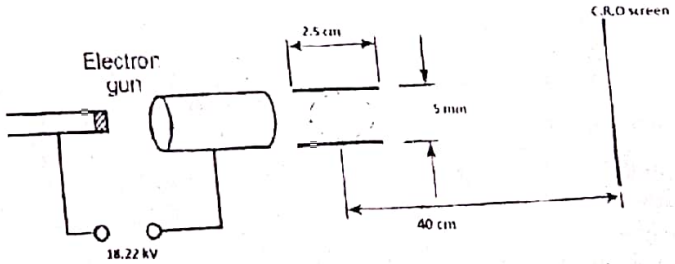
(a) Analyse the information and educate the factory workers about their concerns.

(b) Help the students assess the quality of the factory products and how it monitors the thickness.

ITEM 4:

A group of secondary school students visited one of the Universities in Uganda to take part in science exhibition where rare devices are used to demonstrate core scientific and engineering principles.

The students took a keen interest in one of the system involving various manipulations to of a cathode ray tube to serve various purposes. The instructor operating the system illustrated for the students part of the manipulations as shown in the figure below.



In this manipulation, cathode rays were made to passed midway through parallel plates arranged with a magnetic coil to providing a magnetic field

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of $0.001T$ perpendicular to the electric field between the plates produced by a p.d of $200V$. The students observed a vertical beam on the screen and needed the justification of the whole process and to establish if the beam length was at least $1.5\ cm$ necessary for the required Y-sensitivity of the cathode Ray Oscilloscope (C.RO) screen. The Instructor later made adjustments to bring the Y-sensitivity of the C.R.O screen to exactly $10Vcm^{-1}$. One lecturer brought an a.c source for the instructor to use this CRO to measure. He applied a sinusoidal input from the source to give a steady trace with the time base set so that the electron beam takes $0.01s$ to traverse the screen. The students observed the beam with a total peak to peak height of $4\ cm$ and had 2 complete cycles. The instructor wants to establish the r.m.s voltage and frequency of the input signal and why the time base is necessary.

A second instructor came and made further manipulations to include a metal target embedded in a copper anode after the horizontal plates. The students observed a radiation which the instructor further directed to a metal plate of a negatively charged gold leaf electroscope. They observed a change in the divergence of the leaf. The students group could not understand how the radiation and the deflection were produced. The second instructor informed them that the effect on the metal plate can be used in bank security systems to protect against robbers. The team decided to involve one of the senior six physics student for assistance.

Hint:

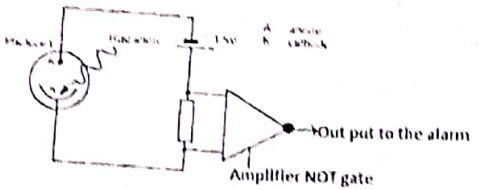
- ✓ Electron mass = $9.11x10^{-31}kg$
- ✓ Electronic charge $e = 1.6x10^{-19}C$

Tasks: You have been selected for help;

(a) Address the needs of the students group and the first instructor.
(b) Establish the operation of the system raised by the second instructor to help banks.

ITEM 5:

A hospital in town has installed an X ray photography room which is kept out of bounds from any intruders. In order to manage this, The Hospital safety officer was told to use a photocell of work function $1.0eV$ connected to the burglar alarm circuit system as shown in the figure below;



However, the safety officer could not establish the wavelength of the radiation which should be incident on the cell to trigger the alarm and he also wondered how the system worked to perform its purpose.

The safety officer wanted to also establish whether the installed X ray machine was producing safe X radiation of frequency within the range of $8.0x10^{16}Hz$ to $3.0x10^{19}Hz$ and made to understand X radiation intensity variation graph with wavelength using the known information of the tube current of $10mA$. It was also known that only 1.0% of the energy of the cathode rays is converted into the X radiation and the rest is generated as heat in the target at a rate of $495W$.

Before obtaining any of the answers, a patient with a swollen neck approached the medical staff which were had moved with the safety officer at $10.00am$. The patient requested for an x ray imaging for her diagnosis but one of the doctors in-charge told her that the correct checkup for the swollen neck does not use X radiation but rather Iodine tracer. The Doctor ordered the Nurse to prepare a radioactive isotope of Iodine $\left(^{131}I\right)$ solution to be injected into the patient to trace using a gamma probe to see if the swelling of the thyroid gland was due to goiter. It was established that the Iodine solution initially has an activity of $6.0x10^{5}Bq$ and the nurse is supposed to inject this solution into the patient at a time when the

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number of atoms have reduced to at most $5.73x10^{11}$. However, the nurse needed assistance to establish this exact time. The nurse also remembered what she was told that the device with a mica window could instead be used to easily trace but she needed to be reminded on how it does so.

As the events were unfolding, electricity went off and the hospital safety officer remembered that they are working on a special future project which can utilize fusion reaction to produce energy for electricity according to the equation below;

$$^{2}_{1}H + ^{2}_{1}H \rightarrow ^{4}_{2}H_e + Energy$$

The officer wants to first understand how binding energy per nucleon graph can be used to account for the energy released and establish the number of reactions and mass of $^{2}_{1}H$ needed to produce $1MJ$ of energy.

Hint:

- Half-life of $^{131}_{53}I$ is 8 *days*
- Electronic charge $e = 1.6x10^{-19}C$
- Planck's constant $h = 6.6x10^{-34}Js$

$Mass\ of\ ^{2}_{1}H = 2.0136U$

$^{4}_{2}H_e = 4.0015u$

$1U = 1.66x10^{-27}kg$

- Speed of light in vacuum $c = 3x10^{8}ms^{-1}$
- Avogadro's' number $Na = 6.02x10^{23}atoms$

Task: As a student of Physics,

(a) Provide the necessary assistance required by the hospital safety officer
(b) Address the concerns of the Nurse.

SECTION B

PART I (ENERGY)

ITEM 6:

A newly established agricultural research center has developed an automated solar-energy harvester for drying crops. The harvester has metal roof of the 25 mm thick which is only blackened at its upper surface. When the crops have fully dried, the center usually checks for the quality by analyzing the sugar content in their extracted juices using of a pair of Polaroids. However, the newly recruited staff lacks clear analytical procedures.
The roof is set to allow sunlight to fall on it normally but the engineering team fears that the resulting surface temperature of the roof puts it at a risk of melting if it exceeds 900K. The team decides to analyze the spectrum of radiation emerging from a narrow hole of a laboratory blackbody cavity maintained at a steady temperature of 2.7 K in order to monitor surface temperature of the roof. They find that its intensity E_λ varies with wavelength, λ as displayed in figure 1.

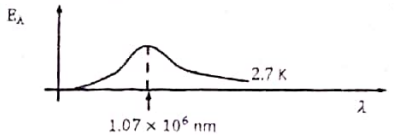


Fig. 1

The team also notes that the peak wavelength of the solar spectrum reaching the upper atmosphere is 500 nm. The team is also worried that the crops may lose their nutrients when the temperature just below the roof of the harvester does not lie in the standard range of drying temperatures. The team has advised the center manager to select a suitable temperature monitoring device. However, the manager doesn't know the criteria for the right selection and how the selected device can be operated.

- Suitable drying temperature lies between 50°C and 80°C.
- Stefan's constant, $\sigma = 5.67 \times 10^{-8} Wm^{-2}K^{-4}$.
- Radius of the sun, $R_s = 6.96 \times 10^{8}$ m.

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- Mean Earth-Sun distance, $r = 1.5 \times 10^{11}$ m.
- Thermal conductivity of metal roof, $k = 0.85 Wm^{-1}K^{-1}$

Tasks:

(a) Provide the staff with analytical procedures and address their concerns together with the fears of the engineering team.

(b) Help the manager to select the right temperature monitoring device and provide operating procedures.

ITEM 7:

A certain school relies on a digital projector operated on power produced by diesel generator. A technician carried out a test run of the generator using 1 mole of oxygen at s.t.p enclosed inside the engine cylinder. He has observed that during the rapid strokes, the gas behaves ideally and swiftly compressed to a volume 0.0166 m^3 before any heat can escape to the surroundings. The technician has been informed that during the process, 350 J is expended by the piston on the gas before its temperature rise to 290 K. He wants to know the cause of temperature rise and is worried that the resulting internal pressure might exceed the engine 1.64×10^5 Pa and cause power disruption.

A projection lantern displayed the following seminar questions:

"Study a progressive wave equation, $y = 0.1 \sin\left(200\pi t - \frac{20\pi}{17}x\right)$, where y is in mm, t is in seconds and x is the distance from a fixed origin in metres and use it to obtain an equation for the resultant wave obtained when the wave is superposed with an identical wave but travelling in opposite direction."

"Design a simple set up of a projection lantern show how it functions to display an image on a screen."

The students have complained that the displayed text is unreadable due coloured edges and lacks clear and sharp focus and could not proceed with the seminar. Before the facilitator address the complaint she wondered how the edges of the text became coloured and how can improve projection lens to ensure color free and how she can adjust the setup for clear focus.

- Standard atmospheric pressure = 1.01×10^5 Pa.
- Density of oxygen at s.t.p = 1.43 kgm^{-3}
- Molar mass of oxygen = 32g

Tasks:

(a) Address concerns of the technician and the facilitator before the students attempt the seminar questions.

(b) Provide the solutions to the seminar questions displayed on the screen.

ITEM 8:

An agricultural research center is developing an automated solar energy harvester for drying crops. The system uses a triangular glass prism of refracting angle 60° to isolate infrared rays from ultraviolet rays. An engineering team sets up the system so that when sunlight falls on the first face, ultraviolet rays pass through the prism symmetrically. The team finds that there are no documentation showing how sunlight is dispersed and wonders whether the specification of the prism makes it suitable for the purpose. The prism is suitable if the net deviation of infrared light does not exceed 10°.

The centre experiences rodents which destroy dried crops inside the storage rooms. The team has integrated an acoustic rodent-repelling track into the solar harvester setup. It detects approaching pest and sends back a scaring signal which propagate in air modeled by a progressive wave equation: $y = 0.02 \sin\left(200\pi t - \frac{40\pi x}{17}\right)$ where x and y are measured in metres, and t is in seconds. The target pests are successfully repelled only if the signal's speed goes beyond 300 ms^{-1}.

The components of the harvester keep drying room at optimal safe temperature by circulating cooling oil through the cooling fins. The standard grade of the oil must possess specific heat capacity lies between 1800 $Jkg^{-1}K^{-1}$ and 2200 $Jkg^{-1}K^{-1}$. The laboratory team performs experimental tests: They heat a 200 g solid to 100°C and quickly dropped into 80 g of specimen oil initially at 15°C, contained in a 120 g calorimeter. The mixture stabilizes at a final temperature of 40°C. One of the team members raises concerns about heat losses of this method and suggests a method in which the liquid continuously flow would be a better alternative.

Hint:

- Refractive index of the prism for ultraviolet light $n_u = 1.50$
- Refractive index for infrared light, $n_i = 1.48$
- Specific heat capacity calorimeter is 400$Jkg^{-1}K^{-1}$

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- Specific heat capacity of solid $450 Jkg^{-1}K^{-1}$

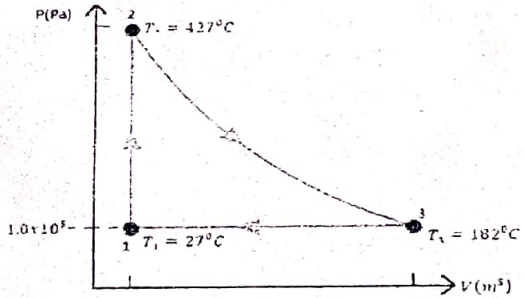
Tasks:

(a) Help the team to address make the necessary documentation, evaluate the suitability of the suitability of the glass prism and determine whether the center succeeds in scaring the rodents.

(b) Assess the standards of the oil in the storage facility and analyse the concerns raised about heat losses.

ITEM 9:

Heat engines are special devices that convert thermal energy into mechanical energy to empower machines in industries and in transport systems. A passenger train uses the heat engine but passengers have been complaining of the slow speed of this train. The train Manager recommended replacement of the engine with another one which can produce at least **160J** of work during one complete cycle process. The one suggested uses **0.3 moles** of an ideal gas around each cycle is shown in the P-V diagram below.



Process **1 → 2** is at constant volume, process **2 → 3** is adiabatic process and process **3 → 1** is at a constant pressure of 1.0×10^5 **Pa**. The train manager does not know if this engine is suitable.

One of the passengers was served with Mango Juice by a train service crew member but this passenger complained bitterly that the juice had a big concentration of sugar which was beyond normal. The train manager wondered how he could determine the true concentration of sugar in the juice to confirm the passenger claims.

At a time when the train was moving **20 ms⁻¹**, the train conductor told the train driver to sound a horn and produce a siren since they were approaching a railway-road junction to prevent any accidents with road vehicles. The progressive sound wave produced had an equation $y = A \sin \pi (960t - \frac{x}{34})$ in air. A police bus carrying traffic officers from a function was approaching the railway-road crossing junction heard the sound from the train horn at a frequency **525 Hz**. The bus driver requested one of the traffic officers aboard to use the speed of this siren sound wave to establish if the bus was travelling at a safe speed to avoid collision with the train.

The traffic officer kept wondering why the train driving crew were not using an optical device known to produce upright images of distant approaching vehicles to avoid any collisions.

HINT:

- ✓ Safe speed at which vehicles approaching the train crossing junction should be almost **54 km h⁻¹**
- ✓ Molar gas constant **R = 8.314 J mol⁻¹ K⁻¹**
- ✓ The ratio of molar heat capacity at constant pressure to that at constant volume for the gas used in the heat engine $\gamma = \frac{5}{3}$

Tasks: As a student of physics, you have been contacted to;

(a) Assist the train manager and the bus driver address their concern.

(b) Establish the design and operation of the device suggested by the traffic officer which can be used to avoid train accidents.

PART II (FIELDS)

ITEM 10:

The internal system of an automated chemical processing plant is monitored by ~~three~~ two reliable sub-units A and B.

Sub-unit A maintains the temperature of a chemical fluid at 373 K in which nichrome heating coil of cross-sectional area, 0.10 mm^2 and rated 36 W, 12 V in series with an ammeter across a d.c supply is immersed. When the sub-unit reduces fluid temperature, the ammeter registers a significantly increased current; the coil becomes hot and gets burnt. The factory manager wishes

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secure another nichrome wire of the same specifications so that in case the wire burns, it can easily be replaced. But before that he wants to understand the cause. He is wondering whether the available 21 cm long nichrome wire of the same size at the same temperature can replace the burnt coil.

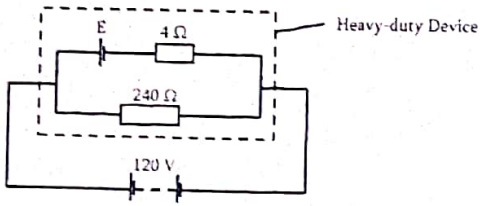


Fig. 2

Sub-unit B consists of a device consisting of rectangular coil of internal resistance 4 Ω connected across a 240 Ω and operated on the main factory 120V d.c supply of negligible internal resistance as shown in figure 2. The device operates on force created on a current carrying conductor placed in a magnetic field. When the supply is switched on, the device pumps the heated chemicals from sub-unit A and sends it to the collection at a steady operational speed. The device draws a total current of 5.5 A from the main factory line. The device is able to provide constant supply of chemicals provided the current through its internal resistance exceeds 4.5A and its back emf, E does not fall below 100 V. The engineering team recommends a device that meets a target of at least 85% efficiency. The manager wants to know how the device can be improved to meet the target and how it operates.

- Temperature coefficient of resistance of nichrome is $8.0 \times 10^{-3} K^{-1}$
- Electrical resistivity at 273 K is 1.08×10^{-6} Ωm.

Tasks:

(a) Analyse sub-units to address the concerns of the plant manager and evaluate the performance of the device.

(b) Help the manager to understand how the device operates and the necessary improvement.

ITEM 11:

During a field inspection, station engineer reveals that two parallel transmission cables, P and Q carry currents of 1.5 A and 3.0 A respectively. Cable P feeds into the station while Q serves as the return path. He notes if the magnetic force per unit length between the wires must remain exceeds 5.0×10^{-6} Nm^{-1}, the cables may result into short circuit. Learners who had visited the station wondered how the force is created with an aim to establish the nature. They were also concerned with minimum separation to prevent the short circuit.

Inside the control room, the engineer presented a sealed filtering module containing a unknown resistor connected in series with 2.0μF capacitor to an a.c source of 32 V(rms), 100 Hz. An a.c meter registers a current of 1.0 mA(rms). The learners notice that as frequency increases, the current rises and levels off at 2.0 mA(rms), but drops to zero when the a.c is replaced d.c supply. The engineer challenges the learners to explain this behavior and identify the components.

The station relies on a heavy-duty emergency backup battery to operate station's emergency systems when the grid fails. For optimal performance of the battery, internal resistance of these cells must fall within the range of 1.5 Ω to 2.2 Ω. The engineer asks the learners carry out an experimental investigation on a single test cell to justify the suitability of the cell. The learners setup a two-metre potentiometer circuit and obtained the following sets of data;

- The balance length with the test cell on an open circuit is 90.6 cm.
- When a 10 Ω shunt is connected across the cell, the balance length shifts to 75.5 cm.

After the experiment the engineer asks them to provide detailed procedure of how to handle the experiment and give the precautions they took to ensure high degree of accuracy while using this device.

- Permeability of free space, $\mu_o = 4\pi \times 10^{-7} Hm^{-1}$

Tasks:

(a) Analyse the information presented and help the learners to address the challenges posed to them by the engineer.

(b) Provide the detailed procedure the student presented to the engineer.

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ITEM 12:

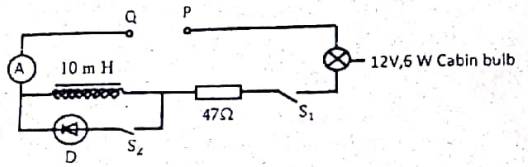


Fig. 3

A technician noticed that a car's interior lighting system was not functioning properly. He suspected that the fault could be in the wiring system and decided to isolate the circuit from the car battery. He inserted a sensitive moving coil current detector, A rated 0.4 A in series with the circuit and device D across the inductor as shown in figure 3. He connected his workshop a.c test generator operating at 14 V (r.m.s), 50 HZ across terminals P and Q. With switch, S_1 closed and S_2 open, neither the cabin bulb nor the moving coil responds properly. He suspected that the inductor's high reactance caused excessive circuit current which would result into burning of delicate coil of the current detector. He is not sure whether his claim is valid. He later decided to close switch, S_1 and to his surprise the bulb immediately illuminated to full brightness. He needs analyse the behavior of the circuit components.

Another technician, Sarah, recommended abandoning the a.c generator and proposed to replacing with a 12 V d.c battery supply, maintaining terminal P positive with respect to Q while keeping S_2 opened. However, is expressed concern that whereas both components may respond properly, cabin bulb may blow due to excessive current. She wants to know the value of a shunt required to protect the bulb during d.c test.

Tasks:

(a) Address the technician's concerns in relation to his power source, and evaluate Sarah's fear with her recommended power source.

(b) Evaluate the response of the components when operated on Sarah's power source, and describe the operation of the current detector.

ITEM 13:

During an electronics club project, members decided to design a radio tuning circuit consisting of a variable capacitor connected in series with a small coil of 200 turns. However, they did not know the exact inductance and required length of the coil. The capacitor consisted of 10 identical pairs of parallel plates connected in parallel. Each pair had an overlapping area of $1.5 \times 10^{-3}m^2$ and a plate separation of 0.50 mm. The circuit successfully received a signal from a local radio station broadcasts at a frequency of 62.0MHz. At this frequency, the current in the tuning circuit was 0.20 A and the coil produced magnetic flux density of $1.0 \times 10^{-3}T$.

They incorporated an LED component for signal indicator into the system to monitor the tuning process. During the tuning, they were surprised to see the LED glowing red when the circuit is completely off the station's frequency, yellow as the tuning knob gets closer and green when correctly tuned to 62.0 MHz. The members could not understand how the component produced light and what brought the variation in coloured lights during the tuning.

Later, the circuit was disconnected, and the capacitor charged with a d.c power supply. The charged capacitor was then reconnected in series with the coil and a meter which detects both a.c and d.c. They did not know why the meter registered constant current. When a solid piece of soft iron was inserted in the coil the members noticed that the iron became significantly warm. The teacher tasked them to investigate the cause of this heating effect and explain its application in modern science and technology.

The following information was provided.

- Permeability of free space, $\mu_o = 4\pi \times 10^{-7}Hm^{-1}$
- Permittivity of free space, $\varepsilon_o = 8.85 \times 10^{-12}Fm^{-1}$

Tasks:

(a) Address the challenges faced by faced by the club members.

(b) Guide the club members to respond to their teachers.

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ITEM 14:

Testing facilities use sensitive meters to measure electricity which are very delicate to handle. A technician is assigned to evaluate the sensitivity of a moving coil galvanometer with an internal resistance 2 Ω. The instrument consists a rectangular coil with 250 turns each of area $4.0 \times 10^{-4} m^2$ suspended in a radial magnetic field of flux density 0.025 T by hairsprings with torsional constant, $3.0 \times 10^{-5} Nmrad^{-1}$. To examine its response limits, the galvanometer is connected in series with a 0.5 m bare constantan wire as a protective resistor. The technician repeatedly charges an air capacitor to 30 V and then discharges it through the galvanometer using a vibrating switch working at 50 times per second. When the circuit is activated, the galvanometer deflects steadily through 0.4 radians. The technician is interested in knowing the energy changes that account for deflection of the galvanometer. He does not know how the wire will eventually heat up under operation. He worries that if the heat generated in the constantan wire reaches 4.55×10^{-2} J, in a span of half a minute after the switch has been activated, the delicate hairsprings may burn out.

The technician decides to re-use the components to assemble a radio receiver tuning circuit designed to receive incoming radio signals at exactly 2.6 kHz. He disconnects the wire from the previous circuit and uses the wire to wind a compact solenoid of length 22.7 cm with 600 turns and a cross-sectional radius 1.42 cm. He connects this solenoid in series with the capacitor and the galvanometer across a 12 V(r.m.s), 40 Hz a.c source. He wants to measure the resonant current through the circuit using the galvanometer, but when he switches on the source, the galvanometer needle seemed not to deflect. He is unable to take readings and does not know why. He wants assistance to understand how the tuning circuit can be modified so that the galvanometer registers a readable current.

Hint:

- Cross sectional area of constantan wire, $A = 4.25 \times 10^{-10} m^2$
- Electrical resistivity of constantan wire, $\rho = 4.9 \times 10^{-8} \Omega m$
- Permeability of free space, $\mu_o = 4\pi \times 10^{-7} Hm^{-1}$

Tasks:

(a) By addressing the concerns of the technician, verify whether the designed circuit will be able to detect the incoming radio signals.

(b) Help the technician to use the galvanometer and estimate the current of the designed circuit.

ITEM 15:

A certain coffee plant operating at 11 kV is to be connected to a power grid at 132 kV. The transformers on the market are at least 95% efficient when operating at an input current of 500 A. The plant manager wishes to secure a brand new transformer but he has been informed that a transformer is suitable for installation if its power losses is not more than 3.32 MW. He wants to know specifications of the required transformer before he buys.

At the plant some other d.c appliance rated 100 V are to be operated from the a.c mains supply and he do not know how to do it.

The frequency of vibration of plant machines is being monitored by a circuit containing an inductor of inductance 32.8 mH connected in series with a resistor of resistance 10 Ω and an a.c meter across an a.c source of peak voltage 10 V. The meter registers root mean square value of current 0.4 A. In order for the machine produce high quality coffee, the frequency of vibration should be maintained at 50.0 Hz. The manager would like to know whether the frequency of the a.c source matches the vibrating frequency of the machine. He has been advised to that in order to monitor the vibration properly, he should buy an appropriate capacitor to connect in series with the available components but he does not know its capacity.

The residents in the neighborhood of the factory often complain about untreated fumes emerging from the factory that their lives are put to risk. They requested the factory manager to help them and install treatment device but the manager could not justify the effect of untreated fumes and he did not know how fume treatment is achieved.

Tasks: Help the manager to

(a) Address the challenges he faces with his coffee plant.

(b) Educate the manager why the residents were concerned and help him to carry out treatment.

THE END

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